MMCL: A Multi-modal Contrastive Learning Framework for Molecular Property Prediction

Mei Gao, Fei Zhu

***Abstract*—Accurately predicting molecular properties can identify more promising drug candidates and facilitate the process of drug discovery. There are advances in methods for molecular property prediction using self-supervised learning. However, most methods tend to focus only on a certain representation, resulting in reduced information diversity. Moreover, less attention is paid to the functional groups in the molecule, which are highly related to the molecular properties. To solve the problem, a multimodal contrastive learning framework called MMCL for molecular property prediction is proposed, which learns common features among similar molecules by utilizing different views of molecules for predicting molecular properties. In addition, by explicitly representing functional groups as nodes in the molecular graph, the model is prompted to learn features related to molecular properties. Prediction results on benchmark datasets and a series of case studies show that MMCL has optimal performance in most property prediction tasks and can capture information related to chemical semantics. Overall, MMCL can be used as an efficient deep learning tool for molecular-related tasks to facilitate the drug discovery process.**

***Index Terms*—Molecular representation, molecular property prediction, contrastive learning, self-supervised learning**

I. Introduction

MOLECULAR property prediction is a key issue in drug discovery [1]. Precisely predicting the properties of molecules helps to identify more promising drug candidates, reduce the average new drug development cost, and accelerate the entire drug discovery process [2]. Computer-aided drug design (CADD), such as quantitative structure-activity relationships (QSAR) and ADME models [3], [4], as an alternative to experimental analysis of molecular properties, reduces the attrition rate of drug development due to poor pharmacokinetics and bioavailability [5]. However, most of them rely on features designed based on expert knowledge [6]. Advances in learning molecular representations directly from observed data using deep learning have attracted a wide range of interest [7]. These methods are trained with various molecular representations as input, including Simplified Molecular Input Line Entry System (SMILES), molecular graphs, etc. Nevertheless, as SMILES is a sequence representation [8], it cannot represent the spatial relationship between atoms, which corresponds to different properties [9]. The molecular graph provides a more direct representation that treats atoms as nodes and chemical bonds as edges, which helps capture the positional relationships in space among atoms in a molecule [9].

Since SMILES sequences resemble natural languages, sequence-based molecular property prediction methods usually adopt methods from the natural language processing domain to embed SMILES sequences. For example, convolutional neural network-based methods [10], [11] process SMILES sequences as fixed-length vectors by padding or truncation, process the input through CNN layers, and add to fully connected layers to predict the result. Methods based on recurrent neural networks (RNNs) [12]–[15] increase model prediction accuracy by improving RNN models. In addition, some studies have considered the key role of functional groups in molecules and developed methods for extracting functional groups hidden in SMILES [9]. These functional groups affect the properties and activity of molecules.

Recently, graph neural networks (GNNs) have shown strong effectiveness in modeling graph-structured data. Essentially, they aggregate neighborhood information to update node characteristics, and can simultaneously capture graph structure and node/edge features. Researchers have proposed many methods for predicting molecular graph properties under supervised learning [16], [17]. However, molecular datasets are highly heterogeneous and relatively small, usually with sparse and incomplete property labels [18]. To overcome this problem, researchers introduced graph self-supervised learning (SSL), which uses extensive unlabeled data to extract intrinsic knowledge via designed pretext tasks. It reduces the reliance on data with labels while also improving the generalization of models. Current graph SSL approaches are generally categorized into two main paradigms: contrast and generation [18]. Contrastive graph SSL methods [19]–[24], [28] construct positive and negative samples through different augmentations, prompting the model to capture common features of similar molecules in different views [19]. The generative graph SSL methods [25]–[27], on the other hand, treat the graph structure or attributes as self-supervision signals. They mask a portion of the graph data and train the model to reconstruct the missing information from the

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remaining context, thereby capturing intrinsic structural and semantic dependencies.

However, most existing methods focus only on extracting molecular features from SMILES sequences or molecular graphs, ignoring the fact that different kinds of inputs can provide richer information than a single representation. Du et al. [29] incorporated messages from two modalities in the prediction of the metabolic stability of molecules, but they simply concatenated the two pieces of information for label prediction. Similarly, Wu et al. [30] also used the aggregated information to utilize the two kinds of information. Although functional groups are highly correlated with molecular properties and activities, only a few methods make use of them in graph SSL. For example, Zhu et al. [31] predicted molecular properties using functional groups together with molecular graphs.

In this study, we present a multimodal contrastive learning framework called MMCL for molecular property prediction, which exploits complementarity in multimodal information to couple molecular graphs and SMILES sequence information. Specifically, MMCL consists of two modules; In the graph-graph contrastive learning module, we adopt the hierarchical molecular graph as the molecular representation, where the motif nodes represent functional groups in the molecule and also serve as a bridge for interactions between lower-order atoms and higher-order molecules. We use attribute masking to obtain augmented views and construct contrastive learning tasks. Besides, we add an attribute prediction task to motivate our model towards recovering node attributes using contextual messages. In the graph-sequence contrastive learning module, a sequence encoder based on a bidirectional gated recurrent unit (BiGRU) is designed to capture molecular features related to structural context and maximize its consistency with graph representation through the contrastive learning tasks. Experimental results show that MMCL is effective in learning the intrinsic knowledge of molecules, and that different views of information play a vital role in the pretraining process. Our major contributions are summarized as follows:

- We proposed a new multimodal contrastive learning framework for learning molecular representations, named MMCL. Specifically, MMCL integrates information in different molecular modes for learning chemical knowledge. Meanwhile, it also considers the important relationship between functional groups and molecular properties.
- We construct a graph-sequence contrastive learning task, where the SMILES sequences and molecular graphs derived from the same molecules were used as positive samples, and the model is trained by maximizing the consistency between positive sample representations, rather than simply concatenating them.
- We conducted comprehensive experiments that indicate that our MMCL performs well in most prediction tasks. In addition, the contribution of each module was evaluated in the ablation experiments, and the results verified that each part plays a vital role in MMCL.

The remainder of this paper is organized as follows. Section II introduced related work. Section III presents the proposed multi-modal contrastive learning framework for molecular property prediction. In Section IV, we conduct a series of comparative studies on benchmark sets of data that illustrate the validity of our model. In Section V, we draw a conclusion.

II. Related Work

A. Graph-based self-supervised learning methods

Due to the small dataset of labeled molecules, supervised models have poor generalization ability in property prediction. Inspired by successful experiences of pretraining in computer vision, researchers have proposed many pre-training models that alleviate the problems associated with label sparsity. Hu et al. [32] set up prediction tasks at the node level and graph level to alleviate the negative migration issue. Similarly, Zang et al. [28] design node-level and graph-level prediction tasks on hierarchical molecular graphs. These methods construct pseudo-labels based on the attributes of the data, which assist in learning molecular representations. However, they rely on the design of the pseudo-label and easily ignore differences in the overall structure of samples.

Methods that follow the contrastive learning framework are divided into cross-scale contrasts and same-scale contrasts [33]. InfoGraph [34] is a cross-scale comparison method that maximizes the mutual information of molecular graphs and their subgraphs. Among the same-scale comparison methods, You et al. [19] designed four graph augmentation methods, using two randomly perturbed versions of the molecular graph as contrastive positive examples. Sun et al. [23] implemented graph augmentation using domain knowledge, considering the global structure of the dataset. However, acquiring such domain knowledge is often costly and limits applicability to large-scale datasets. To mitigate this reliance on expert knowledge, Wang et al. [35] proposed three domain-agnostic augmentation methods for molecular graphs.

Some methods learn graph representations based on graph generation. For example, GraphMAE [25] proposes reconstruction methods for node features, and MGAE [36], and MaskGAE [37] mask and reconstruct paths of graphs.

B. Graphical augmentation methods

We introduce augmentation methods for molecular graphs in this section. In computer vision fields, there are various data augmentations such as geometric transformations, color transformations, and image cropping. However, graph augmentation is not intuitive and difficult to migrate from image augmentation methods [38]. Toward producing high-quality contrast samples, researchers explored various augmentations on graphs, including node dropping, edge perturbation, attribute masking, subgraph, subgraph removal, and substructure substitution.

Node dropping [19] randomly discards nodes and edges connected to them in the graph. Edge Perturbation [19] randomly adds or removes edges from the graph. Typically, the augmentation graphs generated by these two augmentations have a different biological meaning from the original molecule, altering their properties [23]. Some methods [19], [35]removal extract subgraphs and mask subgraphs in a random walk manner, however, which makes it impossible to guarantee that the augmentation graph still represents the main function of the molecule. Substructure substitution replaces valid substructures with bioelectronic bioisomers that will not significantly alter molecular properties, but it requires a deep understanding of the domain. Attribute masking [19] randomly masks the attributes of graph nodes without destroying their structure. Attribute masking does not require domain knowledge, and the augmented graphs show strong semantic correlation with the original graphs.

C. Contrastive Learning Paradigm

Contrastive graph SSL uses the commonality and difference information of sample pairs as supervised signals for learning the prior distribution of the data samples. Contrastive learning avoids the problems of label dependence, pseudo-correlation, and poor model generalization performance in traditional supervised learning, which is widely used in various domains.

In computer vision fields, contrastive learning addresses the problem of learning a priori knowledge distributions with a limited set of labeled datasets. There was no unified paradigm in the early stages of development [39]–[41]. Subsequent works [42]–[46] gradually developed a contrastive learning framework based on negative examples.

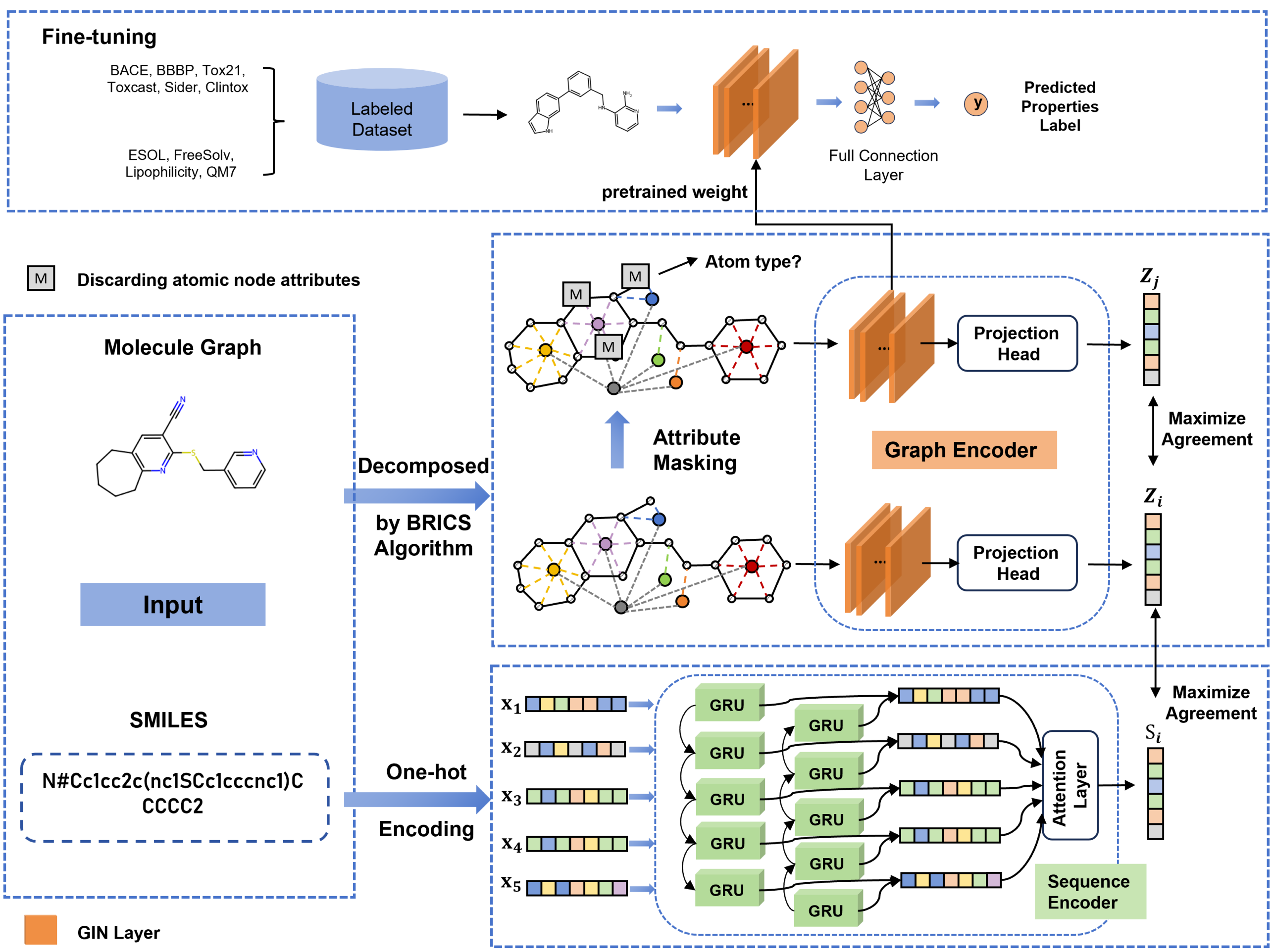

Fig. 1: Overview of the MMCL. MMCL consists of two parts: one is the graph-graph contrastive learning module, and the other is the graph-sequence contrastive learning module. For a molecule, its molecular graph and SMILES sequence are first processed as follows: a hierarchical molecular graph is constructed based on the improved algorithm of BRICS, an enhanced graph is obtained by using attribute masking, and the SMILES sequence is vector-embedded using one-hot encoding. Subsequently, their representations are learned through the graph encoder and sequence encoder, respectively. We set two contrastive tasks to maximize the consistency between different molecular representations, and set an attribute prediction task to enhance the pre-trained model's ability to reconstruct atomic properties.

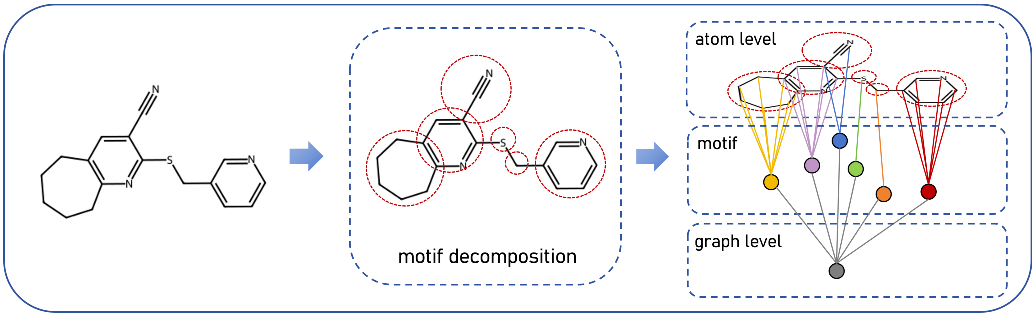

Fig. 2: we first decompose a molecular graph based on the improved algorithm of BRICS, the resulting fragments that usually have chemical properties are added to the molecular graph as motif nodes and connected to the atom nodes that constitute them. A graph-level node is also added to connect these motif nodes.

This framework is implemented as follows: given an image, the semantically identical pairs of positive samples are obtained by data augmentation and mapped to the vector space using a predefined encoder and projection header. The training loss is set to reduce the distance of positive instances and increase the distance of negative instances in the vector space. Researchers also explored multimodal contrast learning, such as clip [47], a pre-training method based on contrasting text-image pairs, which uses text as a supervisory signal to train relocatable visual models.

In recent years, to address the problem of labeled dataset size limitation in molecular property prediction, researchers have proposed a series of pre-training methods [19], [20], [23], [34], [35].

III. Methodology

A. Overview

We propose a multimodal contrastive learning framework called MMCL for molecular property prediction. As shown in Fig. 1, MMCL consists of two components: a graph–graph contrastive learning module and a graph–sequence contrastive learning module. For each molecule, its molecular graph and SMILES sequence are processed by two modality-specific encoders.

Specifically, in the graph–graph contrastive learning module, a hierarchical molecular graph is constructed based on an improved BRICS decomposition to enable interactions between low-order atoms and high-order molecular substructures. An augmented view of the hierarchical graph is then generated via attribute masking, and both the original and augmented graphs are encoded by a graph encoder to learn structure-aware molecular representations. In the graph–sequence contrastive learning module, the SMILES sequence is first transformed into a vector representation via one-hot encoding and subsequently processed by a BiGRU-based sequence encoder. An attention-based aggregation layer is further applied to the BiGRU hidden states to obtain a compact sequence-level representation. The consistency between the graph representation and the sequence representation is maximized through a cross-modal contrastive objective, encouraging the two modalities to capture complementary yet aligned semantic information.

B. Graph-graph Contrastive Learning Module

Molecular graphs naturally display the topological structure of molecules, and these structural features determine the physicochemical properties and biological functions of molecules. Nodes and edges in molecular graphs usually represent atoms and bonds of molecules, respectively. In order to further integrate local and global information in molecules into molecular graphs, we use hierarchical molecular graphs [28] instead of traditional molecular graphs as input, which has been shown to be able to realize the interaction between low-order atoms and high-order molecules, better explore the internal chemical structure of molecules, and help the model learn more comprehensive representations. Subsequently, we design a contrastive learning task based on attribute mask augmentation [19] to learn the representation of compounds.

Specifically, we first decompose the molecular graph into multiple fragments based on the improved algorithm of BRICS, which usually has many chemical properties. These fragments are added to the molecular graph as motif nodes and connected to the atoms that constitute them. In addition, a graph-level node is added to connect these motif nodes. The final hierarchical molecular graph representation is:

$$\begin{aligned}
\mathcal{G} &= (\mathcal{V}, \mathcal{E}) \\
\mathcal{V} &= [\mathcal{V}_{\mathrm{a}}, \mathcal{V}_{\mathrm{m}}, \mathcal{V}_{\mathrm{g}}] \\
\mathcal{E} &= [\mathcal{E}_{\mathrm{a}}, \mathcal{E}_{\mathrm{m}}, \mathcal{E}_{\mathrm{g}}]
\end{aligned} \tag{1}$$

where $\mathcal{V}_{\mathrm{a}}$, $\mathcal{V}_{\mathrm{m}}$, and $\mathcal{V}_{\mathrm{g}}$ represent atomic nodes, motif nodes and graph-level nodes respectively, $\mathcal{E}_{\mathrm{a}}$ represents the chemical bonds between atoms, $\mathcal{E}_{\mathrm{m}}$ represents the edges between motif fragments and the atoms that constitute them, and $\mathcal{E}_{\mathrm{g}}$ represents the edges between graph-level nodes and motif nodes. Fig. 2 illustrates the construction process of the hierarchical molecular graph.

In the pre-training stage, we preprocess the compound into a hierarchical molecular graph as input, and use the graph-level node representation as the molecular representation. In the fine-tuning stage, we assume that the graph encoder has learned the relationship between the internal chemical structure of the molecule and the feature representation through pre-training. Thus, we use a generalized

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approach for obtaining graph-level representations, that is, we do not construct a hierarchical molecular graph, and the molecular representation is obtained by pooling atomic nodes.

In the pre-training stage, after obtaining the hierarchical molecular graph, we designed a contrastive learning task based on attribute masking augmentation for supervised model training. Attribute masking augmentation prompts the model to use context(i.e., remaining nodes)information to restore masked node attributes. Since the structure of a molecule directly affects the properties of the molecule, this augmentation method does not change the structure of the graph, so it has less impact on the properties of the molecule and will not have much impact on model prediction [19]. Specifically, we select a certain proportion of atomic-level nodes from the hierarchical molecular graph $\mathcal{G}$, discard the attributes of these nodes, and thus construct an augmented graph $\mathcal{G}_{\text{aug}}$. We compare the hierarchical molecular graph $\mathcal{G}$ with the augmented graph $\mathcal{G}_{\text{aug}}$ to maximize the consistency of their feature representations. In the pre-training stage, we randomly extract a batch of data, and the contrast loss function of the n-th sample in this batch is defined as follows:

$$\ell_{graph} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{z}_{n,i}, \mathbf{z}_{n,j}\right)/\tau\right)}{\sum_{n'=1,n'\neq n}^{N} \exp\left(\text{sim}\left(\mathbf{z}_{n,i}, \mathbf{z}_{n',j}\right)/\tau\right)} \quad (2)$$

$$\text{sim}(\mathbf{z}_{n,i}, \mathbf{z}_{n,j}) = \frac{\mathbf{z}_{n,i}^{\top}\mathbf{z}_{n,j}}{\|\mathbf{z}_{n,i}\|\|\mathbf{z}_{n,j}\|} \quad (3)$$

where N represents the number of molecules in the current batch, and n' represents the other N-1 samples except n. In our comparison task, the graph $\mathcal{G}$ of the N-th sample and the augmented graph $\mathcal{G}_{\text{aug}}$ are taken as positive sample pairs, and $\mathbf{z}_{n,i}$ and $\mathbf{z}_{n,j}$ are their representation vectors respectively. The graph $\mathcal{G}$ of the n-th sample and the other N-1 augmented graphs $\mathcal{G}_{\text{aug}}$ are taken as negative sample pairs, and $\mathbf{z}_{n',i}$ is the representation vector of the other N-1 augmented graphs $\mathcal{G}_{\text{aug}}$. The cosine similarity function $\text{sim}(\mathbf{z}_{n,i}, \mathbf{z}_{n,j})$ is used to calculate the similarity of the sample pairs.

In addition, in order to enhance the ability of the pre-trained model to reconstruct atomic attributes, we set an atomic prediction task. Specifically, for an augmented graph $\mathcal{G}_{\text{aug}}$ of a molecule, we predict the initial features of the atomic nodes with masked attributes and calculate the cross-entropy loss:

$$\mathcal{L}_{\text{atom}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k}\log(\hat{y}_{i,k}) \quad (4)$$

where $y_{i,k}$ and $\hat{y}_{i,k}$ represent the initial value and predicted value of the feature of the i-th atom, respectively. K represents the number of atom types, and N represents the number of atoms.

We use graph isomorphism network(GIN) as the backbone structure of the graph encoder for feature encoding of graph data [48]. For a molecule, its graph structure is represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, the features of the atomic-level nodes are defined by the atomic attributes, and the fragment-level and graph-level nodes are set to default values. For each node, GIN learns its representation through a message passing framework, that is, aggregating features of neighboring nodes and updating the representation of the current node. Specifically, the representation h_v of the l-th layer node v is defined as follows:

$$h_v^{(l)} = \text{MLP}((1+\epsilon^{(l)})h_v^{(l-1)} + \textstyle\sum_{u\in\mathcal{N}(v)} h_u^{(l-1)}) \quad (5)$$

where $h_v^{(l-1)}$ and $h_u^{(l-1)}$ represent the representation of the l-th layer node v and the neighbor node u respectively, and $\epsilon^{(l)}$ is a learnable weight parameter.

C. Graph-sequence Contrastive Learning Module

We introduce the construction process of the sequence encoder and graph-sequence contrastive learning task in this section. We incorporate information from two modes, namely graph and sequence, into the pre-training framework, and prompt the model to learn the representation of molecules from different modes by setting contrastive learning tasks. In order to capture the contextual information of the sequence, we design a BiGRU-based sequence encoder to learn compound representations. Specifically, we first map the SMILES sequence of the compound to a vector. The SMILES sequence is segmented into characters or character groups, and they are encoded into one-hot vectors, which are combined to obtain the initial vector representation $s_0 \in \mathbb{R}^{m\times d}$ of each sequence, where m and d represent the sequence length (after padding) and the number of character categories, respectively. Subsequently, we input the embedding vector s_0 of the sequence into BiGRU.

Compared with LSTM [49], GRUs [50] simplifies the gate structure and has higher training efficiency. Although Transformer-based sequence models have demonstrated strong capability in modeling long-range dependencies in SMILES strings, we intentionally adopt a BiGRU-based encoder in this work. This design choice is mainly motivated by computational efficiency and scalability in large-scale pre-training, as well as the role of the sequence modality in our framework. Specifically, in the proposed graph–sequence contrastive learning setting, the SMILES sequence is treated as an auxiliary modality to provide complementary semantic information rather than a primary structural representation. Under this consideration, a lightweight yet context-aware sequence encoder is sufficient.

BiGRU consists of a forward GRU and a backward GRU, which can obtain context-sensitive representations by aggregating information from both directions. By modeling the SMILES sequence bidirectionally, BiGRU is able to capture local and medium-range contextual dependencies while maintaining high training efficiency. The hidden state h_t of the characters in the sequence is defined as follows:

$$\begin{aligned} h_t &= \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right] \\ \overrightarrow{h_t} &= \text{GRU}(x_t, \overrightarrow{h_{t-1}}) \\ \overleftarrow{h_t} &= \text{GRU}(x_t, \overleftarrow{h_{t-1}}) \end{aligned} \quad (6)$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ are the hidden states obtained by the forward and backward GRU, respectively, and the current hidden state h_t is calculated by combining $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$.

In the sequence encoder, we incorporate the attention mechanism [51], which has been shown to be effective in identifying molecular substructures. These substructures are usually related to molecular properties and help the model learn features related to molecular properties. Specifically, for a SMILES sequence, we calculate the attention score and summarize the representations of all atoms in a weighted manner to obtain the representation of this sequence z_s:

$$\alpha_{ij} = \operatorname{Softmax}\left[\operatorname{MLP}(h)\right] \tag{7}$$

$$s = \operatorname{FC}\left(\textstyle\sum_{i=1}^{m} \alpha_{ij} h_j\right) \tag{8}$$

where h is the representation of the atom in the sequence, and the attention score α is normalized by the softmax function. The sequence representation is obtained by weighted aggregation of the representations of all atoms.

We construct a multi-modal contrastive task, which is similar to the graph-graph contrastive task. We take the graph and sequence of the n-th sample as the positive sample pair, and the graph of the n-th sample and the sequence of other N-1 samples as the negative sample pair, and calculate the loss as follows:

$$\ell_{gs} = -\log \frac{\exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{s}_{n,j}\right)/\tau\right)}{\sum_{n'=1, n' \neq n}^{N} \exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{s}_{n',j}\right)/\tau\right)} \tag{9}$$

where $\mathbf{z}_{n,i}$ and $\mathbf{s}_{n,j}$ are the representation vectors of the graph $\mathbf{G}$ and sequence $\mathbf{S}$ of the n-th sample respectively, and $\mathbf{s}_{n',j}$ is the representation vector of the other N-1 sample sequences. Finally, the overall loss function is defined as follows:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{atom}} + \mathcal{L}_{\text{graph}} + \lambda \mathcal{L}_{\text{graph\_seq}} \\ &= \mathcal{L}_{\text{atom}} + \textstyle\sum_{\mathbf{g} \in \mathcal{D}} \ell_{graph} + \lambda \sum_{\mathbf{g} \in \mathcal{D}} \ell_{gs} \end{aligned} \tag{10}$$

where λ is a weight coefficient used to weigh the contribution of graph-graph contrastive learning loss and graph-sequence contrastive learning loss.

TABLE I
STATISTICS FOR THE CLASSIFICATION TASK DATASETS (BACE, BBBP, TOX21, TOXCAST, SIDER, CLINTOX) EVALUATED BY ROC-AUC.

Datasets	Molecules	Tasks	Metrics
BACE	1,522	1	ROC-AUC
BBBP	2,053	1	ROC-AUC
Tox21	8,014	12	ROC-AUC
Toxcast	8,615	617	ROC-AUC
SIDER	1,427	27	ROC-AUC
Clintox	1,491	2	ROC-AUC
ZINC15	249,456	-	-

We describe the pre-training and fine-tuning processes in Algorithm 1 and Algorithm 2, respectively. In the

Algorithm 1 Pre-training by MMCL

Input: Molecular graphs $\mathcal{G}_a = (\mathcal{V}_a, \mathcal{E}_a)$ and SMILES $\mathcal{S}$. graph encoder ϕ_g and sequence encoder ϕ_s
Initialize: learningRate is 0.001, batchSize is 256, totalEpoch is 100, augmentation type is AttributeMask, augRatio is 0.2, the number of gnn layers is 5.
Reconstructing the molecular graph G_a through the improved BRICS algorithm: $\mathcal{G}_a \rightarrow \mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = [\mathcal{V}_{\mathrm{a}}, \mathcal{V}_{\mathrm{m}}, \mathcal{V}_{\mathrm{g}}], \mathcal{E} = [\mathcal{E}_{\mathrm{a}}, \mathcal{E}_{\mathrm{m}}, \mathcal{E}_{\mathrm{g}}]$
for *epoch*= 1 to totalEpoch **do**
  for i= 1 to N **do**
    $\mathcal{G}_{\text{aug}} = \text{AttributeMask}(\mathcal{G})$
    $\mathbf{z} = \phi_g(\mathcal{G})$, $\mathbf{z}_{\text{aug}}, \mathbf{y}_{\text{atom}} = \phi_g(\mathcal{G}_{\text{aug}})$
    $\mathbf{s}_0 = \text{One-hot Encoder}(\mathcal{S})$
    $\mathbf{s} = \phi_s(\mathbf{s}_0)$
    $\ell_{graph} = -\log \dfrac{\exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{z}_{n,j}\right)/\tau\right)}{\sum_{n' \neq n}^{N} \exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{z}_{n',j}\right)/\tau\right)}$
    $\ell_{gs} = -\log \dfrac{\exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{s}_{n,j}\right)/\tau\right)}{\sum_{n' \neq n}^{N} \exp\left(\operatorname{sim}\left(\mathbf{z}_{n,i}, \mathbf{s}_{n',j}\right)/\tau\right)}$
    $\mathcal{L}_{\text{atom}} = -\dfrac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,k} \log(\hat{y}_{i,k})$
    $\mathcal{L} = \sum_{\mathbf{g} \in \mathcal{D}} \ell_{graph} + \lambda \sum_{\mathbf{g} \in \mathcal{D}} \ell_{gs} + \mathcal{L}_{\text{atom}}$
  end for
end for

Algorithm 2 Fine-tuning by MMCL

Input: Molecular graphs $\mathcal{G}_a$ with their property labels in baseline datasets $\mathcal{D}$; model ϕ_f.
Load pre-trained weights to initialize ϕ_f
for *epoch*= 1 to *totalEpoch* **do**
  $\hat{y} = \phi_f(G_a)$
  if D is the dataset for classification task **then**
    $L_{fintune} = \frac{1}{N} \sum_{i=1}^{N} [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$
  else if D is the dataset for regression task **then**
    if D in [QM7,QM8,QM9] **then**
      $L_{fintune} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i|$
    else if D is the dataset for regression task **then**
      $L_{fintune} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$
    end if
  end if
end for

TABLE II
STATISTICS FOR THE REGRESSION TASK DATASETS (ESOL, FREESOLV, LIPOPHILICITY, QM7).

Datasets	Molecules	Tasks	Metrics
ESOL	1,128	1	RMSE
FreeSolv	643	1	RMSE
Lipophilicity	4,200	1	RMSE
QM7	7,165	1	MAE

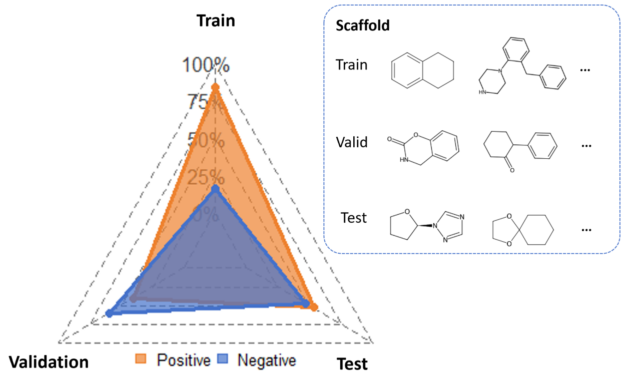


Fig. 3: Results of BBBP split by molecular scaffold. The ratio of training set, validation set and test set is 8:1:1. Each set shows the scaffolds of some molecules.

TABLE III
TABLE OF PARAMETERS IN THE PRE-TRAINING PHASE.

Parameters	Value
Epoch	100
Batch Size	256
Num Workers	8
Optimizer	Adam
Learning Rate	1×10^{-3}
Graph Augmentation Type	Attribute Masking
Graph Augmentation Ratio	0.2
Graph Encoder	GIN
Layers of GINs	5
Sequence Encoder	BiGRU
Embedding Dimension	256
Loss Weight(λ)	0.1

TABLE IV
TABLE OF TRAINING PARAMETERS FOR THE BASELINE MODELS.

Parameters	Value
Epoch	100
Batch Size	256
NumWorkers	8
Optimizer	Adam
Learning Rate	1×10^{-3}
Graph Encoder	GIN
Layers of GINs	5
Embedding Dimension	300

pre-training stage, the model constructs a hierarchical molecular graph $\mathcal{G}$ (line 3 of Algorithm 1) and performs attribute mask augmentation on the molecular graph to obtain the augmented graph $\mathcal{G}_{\text{aug}}$. Afterwards, the vector representations $\mathbf{z}$ and $\mathbf{z}_{\text{aug}}$ of the graph $\mathcal{G}$ and $\mathcal{G}_{\text{aug}}$ are obtained through the graph encoder ϕ_g (lines 6 to 7 of Algorithm 1). The model encodes the sequence $\mathbf{S}$ with one-hot coding and obtains vector representation $\mathbf{s}$ through the sequence encoder ϕ_s (lines 8 to 9 of Algorithm 1). The loss of the self-supervised tasks is calculated according to the description in Section III-B and Section III-C (lines 10 to 13 of Algorithm 1).

In the fine-tuning phase, the model is initialized using the weights obtained from the pre-training (line 2 of Algorithm 2), and then the input molecules are mapped to the vector space (line 4 of Algorithm 2), which are applied to different molecular property prediction tasks (lines 5 to 13 of Algorithm 2).

IV. Experiments and Analysis

In this section, we first present the data and settings of the experiments, and then verify the performance of our proposed pre-training framework by comparing it with the baseline methods, carrying out ablation experiments, and related studies.

A. Datasets

We use the ZINC15 dataset [52] for pre-training, which contains about 2.49×10^5 unlabeled small molecule compounds. We use the SMILES sequence and molecular graph of each small molecule as input. To ensure consistent molecular representations, all SMILES strings from both pre-training and downstream datasets are standardized before being used as model inputs. The standardization includes structure cleaning and unified expression of stereochemistry, charges, and aromaticity, preventing multiple forms of the same molecule from affecting learning. For fine-tuning, we use six datasets selected from Moleculenet [53] for classification tasks, namely BACE, BBBP, Tox21, Toxcast, SIDER, and Clintox. We use four datasets for regression tasks, namely ESOL, FreeSolv, Lipophilicity, and QM7. The molecules in these datasets have property labels. Table I and II give the statistics of these datasets.

We split the benchmark dataset based on the skeleton of the compound with a ratio of 8:1:1, ensuring that the training, validation, and test sets have molecules with different skeletons. Figure 3 shows the split results of BBBP and its molecular skeleton. Intuitively, the training set is unbalanced, which poses a greater challenge to the generalization ability of the model compared to the random partitioning method.

To ensure consistent molecular representations, all SMILES strings from both pre-training and downstream datasets are standardized before being used as model inputs. The standardization includes structure cleaning and unified expression of stereochemistry, charges, and aromaticity, which reduces inconsistencies caused by different SMILES representations of the same molecular graph.

TABLE V
RESULTS FOR CLASSIFICATION TASKS OF MOLECULAR PROPERTY PREDICTION. THE MEAN (AND STANDARD VARIANCE) OF THE ROC-AUC (%) FOR 10 RANDOM SEEDS ON THE SIX DATASETS IS REPORTED. THE BEST AND SECOND BEST RESULTS ARE SHOWN BY BOLDING AND ADDING UNDERLINES, RESPECTIVELY. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Models / Datasets	BACE	BBBP	Tox21	Toxcast	SIDER	Clintox	Average
ContextPred	78.8 (1.6)	70.1 (0.9)	<u>74.5</u> (0.9)	63.0 (0.4)	59.9 (1.1)	73.2 (3.8)	69.9
Infomax	73.8 (1.3)	67.2 (1.3)	74.4 (0.9)	63.1 (0.5)	58.5 (0.9)	73.0 (4.2)	68.3
EdgePred	80.1 (2.2)	70.2 (1.5)	73.6 (0.6)	63.0 (0.6)	58.6 (1.3)	75.5 (2.4)	70.2
AttrMasking	75.5 (1.9)	67.9 (1.3)	73.8 (0.7)	64.0 (0.3)	58.6 (0.9)	<u>85.1</u> (2.4)	70.8
GraphCL	73.5 (1.8)	70.1 (0.9)	73.7 (0.7)	63.3 (0.7)	59.6 (0.5)	79.2 (1.9)	69.9
JOAO	73.0 (1.1)	68.9 (1.1)	73.8 (0.7)	63.1 (0.4)	60.3 (1.0)	83.5 (1.9)	70.5
JOAOv2	72.2 (1.4)	68.8 (1.1)	74.2 (0.5)	63.4 (0.5)	59.7 (0.6)	83.7 (2.0)	70.3
MoCL	68.6 (1.3)	65.9 (1.1)	69.7 (0.5)	60.9 (0.3)	57.3 (0.5)	57.7 (1.4)	63.3
GraphMAE	80.2 (1.2)	68.3 (1.1)	74.3 (0.6)	62.7 (0.1)	59.7 (0.6)	80.3 (3.3)	70.9
SimGEACE	77.8 (0.7)	66.5 (1.0)	73.9 (0.7)	63.1 (0.4)	58.9 (0.5)	58.7 (1.2)	66.5
MolCLR	78.5(1.3)	**70.8** (0.8)	71.6(0.8)	62.7(1.9)	59.7(0.4)	**86.9** (2.3)	<u>71.7</u>
HiMol	<u>81.2</u> (0.9)	66.9 (2.1)	**77.1** (0.3)	<u>65.8</u> (0.6)	**60.7** (0.4)	68.6 (2.7)	70.1
MMCL (Ours)	**82.6** (1.0)	<u>70.3</u> (0.7)	<u>75.9</u> (0.6)	**66.7** (0.5)	<u>60.4</u> (0.5)	75.2 (1.8)	**71.9**

B. Experiment Settings

We use the Adam optimizer [54] in both pre-training and fine-tuning. The parameters of the pre-training phase are as follows. The model training iterations are 100, the batch size is 32, the learning rate is set to 0.001, and the weight in the loss function is 0.1. The graph encoder uses a 5-layer GIN (hidden layer dimension is 256) as the backbone structure, and the attribute mask rate is 0.2. The sequence encoder uses BiGRU for encoding. Table III lists the relevant parameter settings of the model.

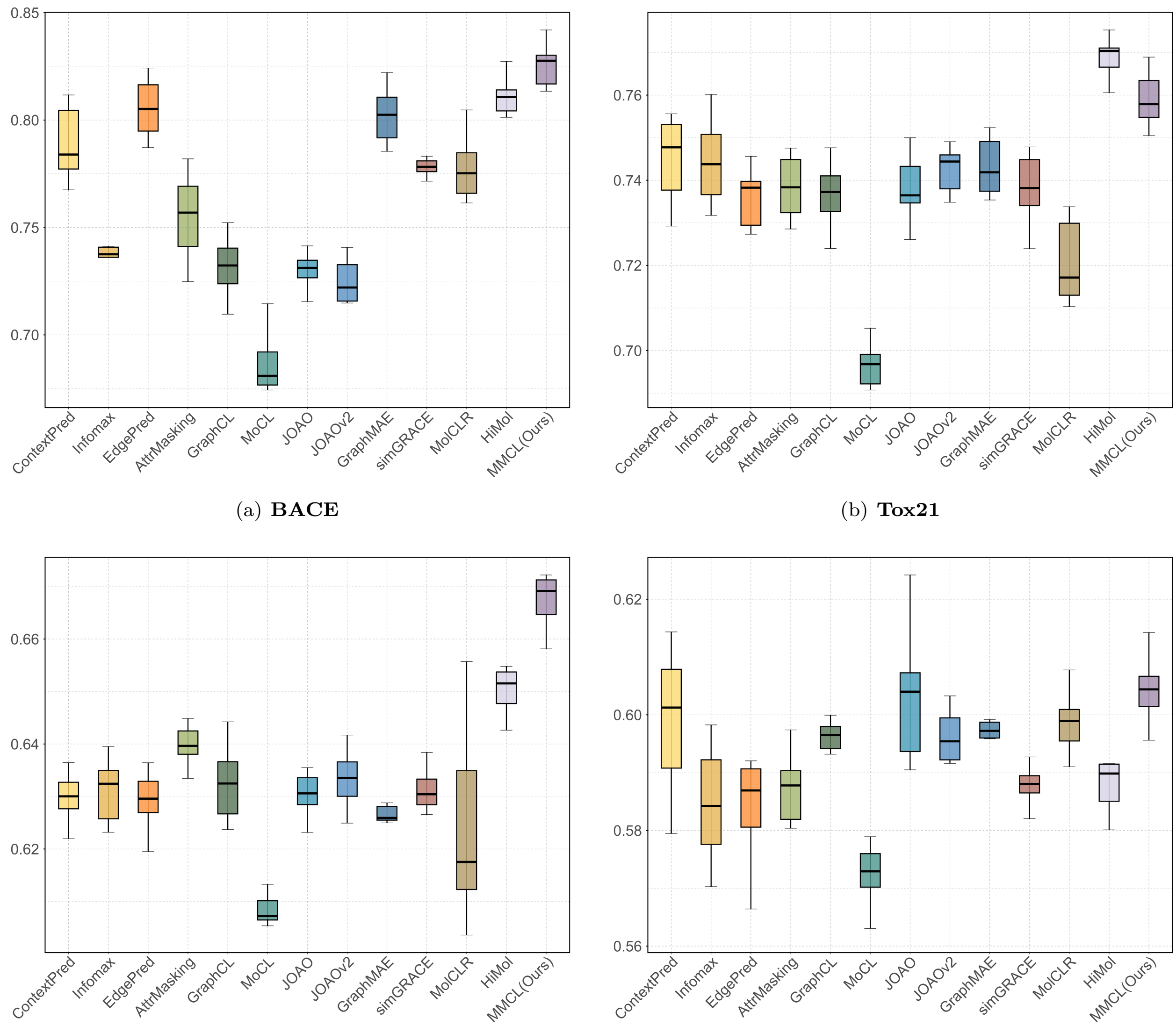
C. Molecular Property Prediction

Baseline Models We compared MMCL with the most advanced methods for molecular property prediction, including prediction-based methods: ContextPred, InfoMax , EdgePred, AttributeMask, and ContextPred [32]; generative reconstruction-based methods: GraphMAE [16] and HiMol [55]; and contrast task-based methods: GraphCL [19], MoCL [23], JOAO [20], JOAOv2 [20] SimGRACE [24] and MolCLR [56]. Since MoCL requires high computing power to calculate the similarity matrix on the ZINC15 dataset, we use the approach given by the official code of MoCL for pre-training. Other models were pre-trained on the ZINC15, and experimental parameter settings are listed in Table IV.

Classification Tasks We evaluated the effectiveness of MMCL for molecular property prediction by performing classification tasks across six datasets. Table V reports the mean and standard deviation of the ROC-AUC (%) results in the classification tasks. We summarize the results analysis as follows. We achieved strong performance on five of the six datasets. In particular, we obtained an improvement of 1.4% on BACE and 0.9% on Toxcast compared to the best-performing baseline, and achieved the highest overall average performance across all datasets. The experimental results demonstrate the effectiveness of our framework in learning relevant chemical knowledge within molecules. In addition, although we have achieved the best results in average performance, the results on the Clintox show that our model still has room for further improvement.

We further analyze the results of the classification task through box plots. The observations in Fig. 4 are summarized as follows. (1) The median of the MMCL box is higher than that of other baselines in three datasets, indicating that its overall performance is better. (2) The box range (i.e., the distance from the lower quartile to the upper quartile) and whisker length (i.e., the distance between the minimum value and the lower quartile, and the maximum value and the upper quartile) of MMCL are relatively small, indicating that the prediction results are more consistent and the model has better stability. (3) On the datasets BACE, and Toxcast, more than half of the ten results of MMCL are completely higher than the best result of the second-best baseline model. Specifically, in bace, six of MMCL's results are higher than the top result of GraphMAE. In Tox21, seven of MMCL's results are higher than the top result of ContextPred. In particular, all ten results of MMCL on Toxcast are better than the top result of AttrMasking.

Regression Tasks Table VI reports the mean and standard deviation of the RMSE/MAE results in regression tasks. Based on MoleculeNet [53], we evaluate it using root mean square error (RMSE) on ESOL, FreeSolv, and Lipophilicity datasets, and mean absolute error (MAE) on the QM7 dataset. The results show that we achieved the best or second-best results on three of the datasets. In



(a) **BACE** (b) **Tox21** (c) **Toxcast** (d) **SIDER**

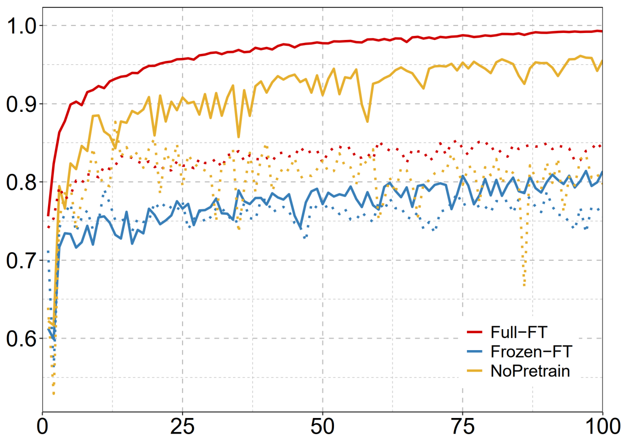
Fig. 4: Box plots of the results in the classification task. Boxes of the same color correspond to the same model. The horizontal axis is the model name and the vertical axis is the ROC-AUC result.Fig.4 (a), (b), (c), and (d) are the results on the datasets BACE, Tox21, Toxcast, and SIDER, respectively.

particular, on the FreeSlov dataset, the results are reduced by 9.32% compared to the second-best baseline.

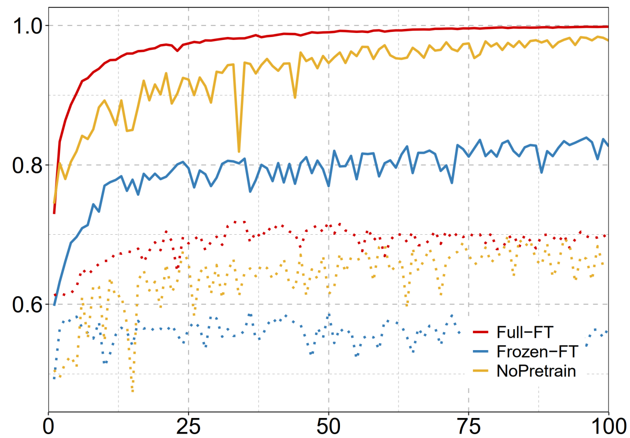
It is worth noting that MMCL exhibits relatively weaker performance on the QM7 dataset among the regression tasks. The QM7 dataset focuses on quantum chemical properties, which are highly sensitive to precise atomic coordinates and three-dimensional molecular conformations. In contrast, MMCL primarily models molecular topology through hierarchical graph structures and explicit functional group representations, without incorporating explicit three-dimensional geometric information. As a result, hierarchical motif-based representations are more suitable for tasks dominated by functional group interactions and global structural patterns, while they may be less effective for quantum properties that depend on fine-grained geometric and electronic details.

D. Ablation Experiment

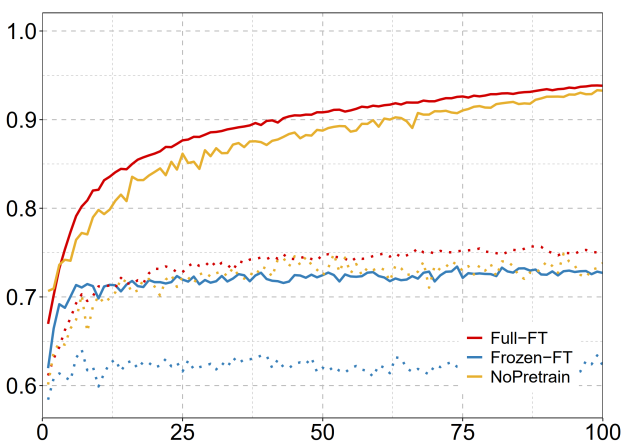
We performed ablation studies to evaluate the contribution of each component of MMCL. Specifically, for “w/o graph-seq cl”, we only retain the graph-graph contrastive learning module, and the loss function is $\mathcal{L} = \mathcal{L}_{\text{atom}} + \mathcal{L}_{\text{graph}}$. For “w/o graph-graph cl”, we only retain the graph-sequence comparison module, and the loss function is $\mathcal{L} = \mathcal{L}_{\text{graph\_seq}}$. Since attribute masking is only used for molecular graphs, the attribute prediction loss is not included in the loss function. In addition, we introduce a hierarchical molecular graph into our framework to realize the interaction between low-order atoms and high-order molecules during training. Therefore, we also set up a w/o hierarchical mol-graph ablation experiment, where the hierarchical molecular graph is not constructed during



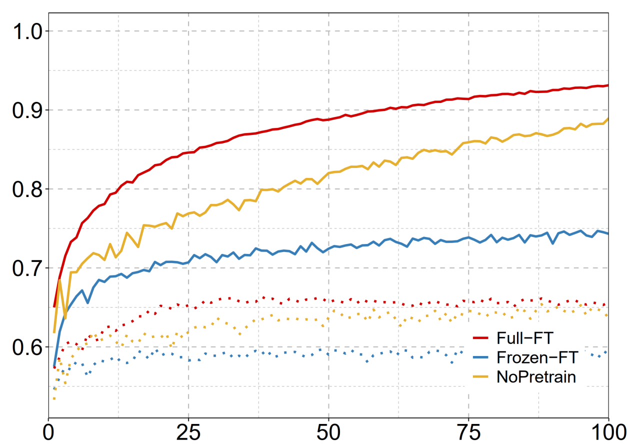
(a) **BACE**



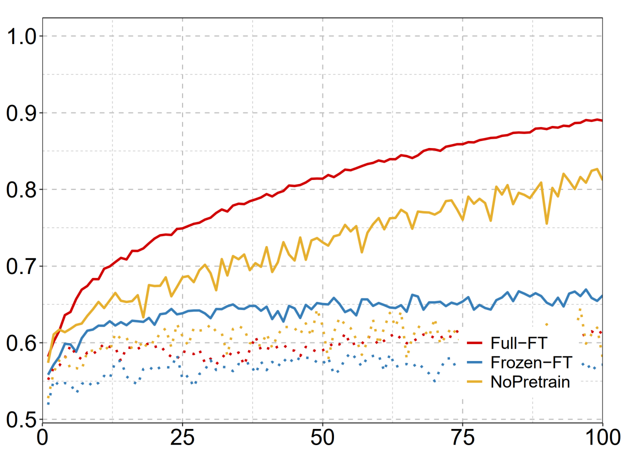
(b) **BBBP**



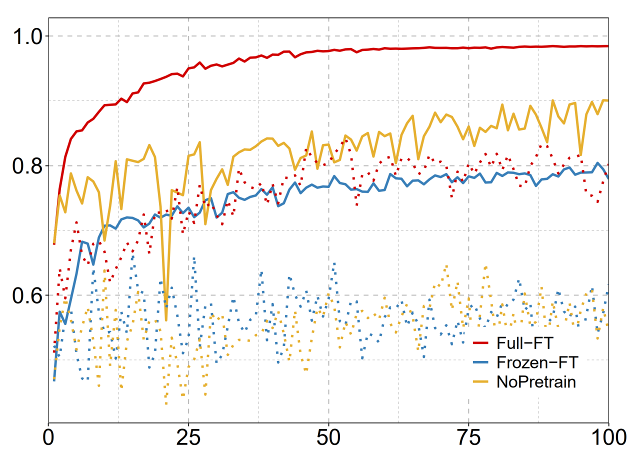
(c) **Tox21**



(d) **Toxcast**



(e) **SIDER**



(f) **Clintox**

Fig. 5: The ROC-AUC curves of different fine-tuning strategies on the training set and test set of classification tasks. The horizontal axis represents the iteration number, and the vertical axis represents the ROC-AUC values. The curves of the two fine-tuning strategies based on pre-trained weights (Full-FT and Frozen-FT) are more stable, proving that the pre-training framework can effectively train the parameters of the graph encoder and use the knowledge for downstream tasks. In addition, the full fine-tuning mode (Full-FT) achieved the highest ROC-AUC value, indicating that the model with general knowledge can capture more detailed features on specific tasks.

TABLE VI
RESULTS OF THE REGRESSION TASK FOR MOLECULAR PROPERTY PREDICTION. WE REPORT THE MEAN (AND STANDARD DEVIATION) OF RMSE/MAE ON ESOL, FREESLOV, LIPOPHILICITY, AND QM7. THE EVALUATION METRIC FOR ESOL, FREESLOV, AND LIPOPHILICITY IS RMSE; THE EVALUATION METRIC FOR QM7 IS MAE. THE BEST AND SECOND-BEST RESULTS ARE BOLDED AND UNDERLINED, RESPECTIVELY. SMALLER VALUES INDICATE BETTER PERFORMANCE.

Models / Datasets	ESOL	FreeSolv	Lipophilicity	QM7
ContextPred	1.428 (0.053)	3.351 (0.153)	0.774 (0.016)	252.939 (23.278)
Infomax	1.403 (0.055)	4.221 (0.573)	0.756 (0.015)	223.300 (21.209)
EdgePred	1.479 (0.057)	3.883 (0.597)	0.756 (0.018)	210.331 (26.332)
AttrMasking	1.257 (0.027)	3.982 (0.474)	0.771 (0.012)	198.953 (18.328)
GraphCL	1.331 (0.025)	4.674 (0.753)	0.760 (0.017)	209.452 (27.890)
JOAO	1.402 (0.056)	5.178 (0.790)	0.759 (0.013)	212.261 (17.950)
JOAOv2	1.321 (0.054)	6.983 (1.444)	0.761 (0.011)	228.178 (27.719)
MoCL	1.698 (0.035)	3.319 (0.058)	1.026 (0.008)	162.451 (1.865)
GraphMAE	1.321 (0.040)	3.299 (0.321)	0.760 (0.017)	223.480 (25.077)
SimGEACE	1.593 (0.054)	3.464 (0.486)	0.854 (0.028)	97.831 (5.423)
MolCLR	1.297 (0.033)	3.237 (0.293)	**0.740** (0.014)	**74.983** (8.651)
HiMol	**0.937** (0.024)	<u>2.747</u> (0.058)	0.791 (0.012)	<u>75.571</u> (1.031)
MMCL (Ours)	<u>1.163</u> (0.062)	**2.491** (0.132)	<u>0.779</u> (0.018)	168.903 (8.196)

pre-training.

Table VII presents the results of the ablation experiments. Removing any component of MMCL leads to performance degradation on benchmark datasets, indicating that different modalities contribute to molecular representation learning.

Among the considered components, introducing the hierarchical molecular graph consistently improves performance across datasets. By explicitly incorporating motif-level nodes derived from BRICS decomposition, the hierarchical graph enables interactions between atom-level representations and higher-order substructures. These motif nodes correspond to functional groups, which are closely related to molecular properties. The results suggest that such explicit modeling of functional groups provides particularly informative structural cues and plays a crucial role in learning discriminative molecular representations.

In addition, the two contrastive learning variants, namely graph–graph and graph–sequence contrastive learning, exhibit comparable effects when used individually. This observation indicates that the SMILES-based sequence modality serves as a complementary view rather than a standalone representation. By encouraging consistency between graph and sequence representations, the multimodal contrastive objective helps regularize the learned embeddings and enrich the semantic information captured by the graph encoder.

These results demonstrate that MMCL benefits from both hierarchical structural modeling and multimodal contrastive learning. While the explicit modeling of functional groups through hierarchical graphs appears to be the most influential factor, the incorporation of sequence information further enhances representation robustness and generalization.

E. Fine-tuning Strategy

The effect of different fine-tuning strategies on downstream tasks was explored. In both the full fine-tuning mode (Full-FT) and the frozen mode (Frozen-FT), the model loads the pre-trained weights. The difference is that the full fine-tuning mode trains all weights in the graph encoder and the property prediction classifier during fine-tuning, while the frozen mode only trains the weights in the classifier. We also adopted the from-scratch training mode (NoPretrain), in which the graph encoder does not load the pre-trained weights during fine-tuning. The running results of different fine-tuning strategies are shown in Fig. 5. We analyze the results as follows: (1) The training and test curves of the full fine-tuning mode and the frozen mode have better stability, proving that MMCL can effectively train graph encoders and apply the learned knowledge in fine-tuning. (2) The full fine-tuning mode converges to the best performance faster than the from-scratch training mode, indicating that the pre-trained weights provide rich general knowledge, and the model can capture more detailed features by fine-tuning specific tasks based on this knowledge. (3) The training results of the frozen fine-tuning mode show that general features cannot be fully applied to specific tasks, resulting in the encoder being unable to better extract deep information for specific tasks.

F. Molecular Representations Visualization

We visualize molecular representations in the test set learned by MMCL as shown in Fig. 6. We choose BACE

TABLE VII

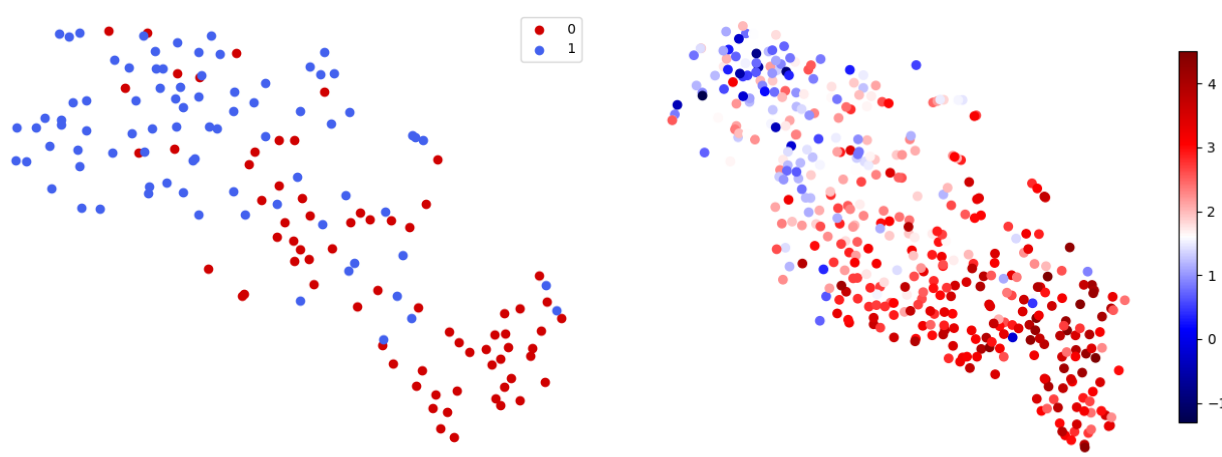
RESULTS OF ABLATION EXPERIMENTS ON CLASSIFICATION DATASETS. THE MEAN (AND STANDARD VARIANCE) OF THE ROC-AUC (%) FOR 10 RANDOM SEEDS ON THE SIX DATASETS IS REPORTED.

Models / Datasets	BACE	BBBP	Tox21	Toxcast	SIDER	Clintox	Average
w/o graph-seq cl	79.6 (1.1)	70.2 (1.7)	75.5 (0.5)	65.5 (0.4)	59.0 (1.4)	73.8 (2.3)	70.6
w/o graph-graph cl	79.4 (1.2)	68.7 (1.4)	75.7 (0.8)	65.9 (0.6)	59.9 (0.8)	74.4 (2.4)	70.7
w/o hierarchical mol-graph	77.4 (1.2)	68.4 (1.0)	75.4 (0.5)	65.5 (0.5)	59.2 (1.3)	74.7 (3.6)	70.1
MMCL (Ours)	**82.6** (1.0)	**70.3** (0.7)	**75.9** (0.6)	**66.7** (0.5)	**60.4** (0.5)	**72.5 (1.8)**	**71.9**

from the classification task and Lipophilicity from the regression task, and use t-SNE [57] to project the molecular representations into two-dimensional space. Different colors in the figure represent different molecular property labels. The figure shows that the molecular representations are grouped according to the labels. Specifically, the property labels of BACE represent the binding results of inhibitors of human β-secretase 1 (BACE-1) [58]. In Fig. 6 (a), the two types of molecules tend to be at opposite ends of the spectrum. Lipophilicity provides the octanol/water distribution coefficient (logD at pH 7.4) of the compound, which affects the permeability and solubility of the compound [53]. In Fig. 6 (b), the exp value of molecules increases gradually from the top left to the bottom right. The results show that our model can learn features related to the properties of the molecule, proving the effectiveness of MMCL.

V. Conclusion

In this study, a multimodal contrastive learning framework MMCL for molecular property prediction is presented, which exploits the complementarity of multimodal information to couple molecular graphs and SMILES sequence information. Specifically, MMCL consists of two modules: (1) A graph-graph contrastive learning module, which introduces hierarchical molecular graphs to explicitly represent functional groups and perturbs node features with attribute masking in the graph augmentation strategy. (2) The graph-sequence contrastive learning module obtains a SMILES representation with structural context through a sequence encoder, and maximizes the feature consistency with the graph representation. Experiments on benchmark datasets show that MMCL has optimal performance in most property prediction tasks, while ablation experiments validate the effectiveness of modules in MMCL. We also perform visual analysis which shows that the graph encoder trained by MMCL can capture chemically relevant semantic information and has a stronger generalization ability. Although MMCL has the best average performance on property prediction tasks, it still faces some limitations, such as insufficient ability to capture relationships within molecular structures related to lipophilicity and clinical toxicity. Additionally, since SMILES representations can occasionally introduce



(a) **BACE** (b) **Lipophilicity**

Fig. 6: Visualization results of molecular representations on the molecular properties test set. Fig.6(a) is the result of the classification task BACE. Fig.6(b) is the result of the regression task Lipophilicity, with increasing values of exp from top left to bottom right.

errors when encoding molecular sequences, future work will explore more robust alternatives, such as SELFIES, to improve molecular representation reliability [59]. In future research, relevant influencing factors will be further analyzed to optimize the pretraining framework. Overall, MMCL can serve as an efficient deep learning tool for molecular-related tasks, facilitating the drug discovery process.

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